

SATNET EDU / Lecture 02

Orbits, Constellations, and Coverage

How satellite motion determines where and when access is possible.

LECTURE 02

From one moving satellite to coverage over time

Predict the motion. Construct the constellation. Check the coverage.

Instructor: Yi Ching (David) Chou

Altitude Sets the Pace of a Circular Orbit

A satellite keeps moving while gravity bends its trajectory around Earth.

► Gravity bends the path.

- Velocity points along the orbit. Gravity pulls toward Earth's center, continuously changing the direction of motion.

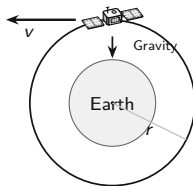
► Use distance from Earth's center.

- Orbital radius is $r = R + h$, where R is Earth's radius and h is altitude. Altitude alone is not the orbital radius.

► Higher circular orbits take longer.

- A larger circumference and lower speed mean fewer revolutions per day.

Motion along the orbit, gravity inward



$$v = \sqrt{\mu/r}$$

$$T = 2\pi\sqrt{r^3/\mu}$$

v : orbital speed
 T : orbit period
 μ : Earth's gravity parameter

Altitude	Speed	Period
550 km	7.59 km/s	95.5 min
1200 km	7.26 km/s	109.3 min

Ideal circular orbits. $R = 6371$ km, $\mu = 398600.44$ km³/s². Diagram is not to scale.

Inclination Controls the Latitudes Reached

Altitude sets the orbit's size. Inclination tilts its plane relative to the equator.

► **Inclination is the orbital tilt.**

- $i = 0^\circ$ is equatorial. At $i = 90^\circ$, the orbit passes over both poles. We compare prograde orbits with $0^\circ \leq i \leq 90^\circ$.

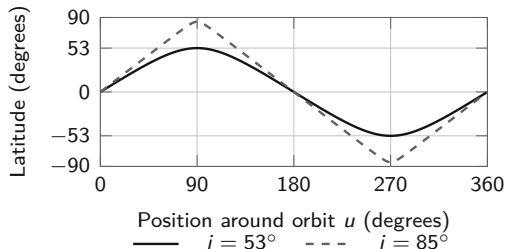
► **The point below a satellite moves.**

- This **subsattellite point** reaches latitudes from $-i$ to $+i$. A 53° orbit never passes directly above 70° N.

► **Radio coverage extends farther.**

- A terminal can see a satellite away from overhead. The usable footprint extends beyond the ground track, as we calculate next.

Latitude during one revolution



$$\sin \phi = \sin i \sin u$$

ϕ : subsatellite latitude

u : angle from the northbound equator crossing

Spherical Earth and circular orbits. These curves show subsatellite latitude, not a radio-coverage map.

Earth Rotates Beneath the Orbital Plane

A ground track is the path of the subsatellite point on the rotating Earth.

► **An orbit is not a fixed map curve.**

- In this ideal model, the orbital plane is fixed in space. Earth rotates eastward underneath it.

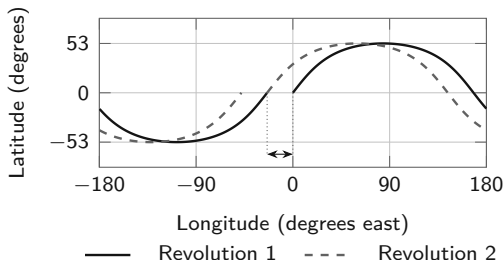
► **The next crossing shifts west.**

- Earth turns 23.9° during a 95.5-minute orbit. The next northbound equator crossing shifts west by that amount.

► **Orbit period is not revisit time.**

- After one orbit, the satellite is over a different longitude. Revisiting a site depends on both motions.

Two revolutions of the same 53° orbit



$$\text{Westward shift} \approx 360^\circ \times \frac{95.5}{1436.1} = 23.9^\circ$$

550 km circular orbit, no precession. Earth's sidereal rotation period is 1436.1 min. Curves break at the map's date line.

Planes, Spacing, and Phasing Shape the Constellation

The same number of satellites can be distributed across orbital planes in different ways.

► **Planes spread orbit orientations.**

- Rotating planes about Earth's axis spreads their equator crossings across different longitudes.

► **Spacing sets the gap within a plane.**

- With S evenly spaced satellites, the angular gap is $360^\circ/S$. More planes at fixed total count leave fewer satellites per plane.

► **Phasing staggers satellite positions.**

- Offsetting neighboring planes changes when their satellites arrive. Satellite count alone cannot describe coverage.

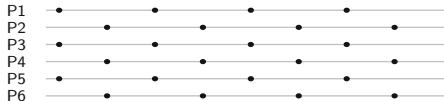
Two layouts, each with 24 satellites

3 planes $\times$ 8 satellites: 45° spacing



Alternate rows are shifted by half a slot

6 planes $\times$ 4 satellites: 90° spacing



Rows wrap around. Horizontal position is orbital angle.

Schematic layouts at a common altitude and inclination. Half-slot offsets illustrate phasing, not a universal constellation rule.

The Elevation Mask Sets the Usable Footprint

For one satellite, coverage is the set of ground locations that meet the chosen access conditions.

► **A mask defines a circular footprint.**

- An elevation mask defines a circle centered on the subsatellite point. Its radius is measured along Earth's surface.

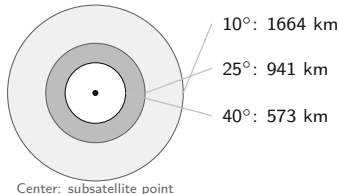
► **A higher mask makes it smaller.**

- A steeper viewing angle excludes distant contacts. At 550 km, a 25° mask gives a radius of about 941 km.

► **Geometry is one access condition.**

- Antenna steering, blockage, and capacity can restrict service further. A geometric footprint does not guarantee throughput.

A stricter mask removes the outer region



$$\psi = \arccos\left(\frac{R}{R+h} \cos e_{\min}\right) - e_{\min}$$

Surface radius $\rho = R\psi$

R : Earth radius h : altitude $e_{\min}$: mask
 ψ : angle at Earth's center, in radians for $R\psi$

550 km altitude and spherical Earth, $R = 6371$ km. Circles compare surface radii, not commercial service beams.

Contact Duration Depends on More Than Altitude

A ground terminal has a contact while the satellite stays above its elevation mask.

► **A contact has a start and an end.**

- Elevation rises to a peak and falls again. The two crossings of the mask define the usable contact window.

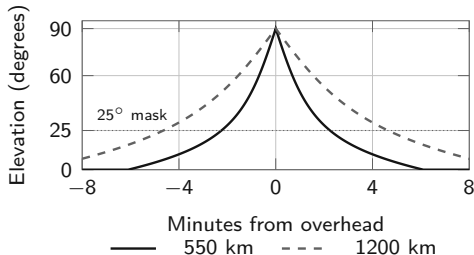
► **Higher orbits can prolong contact.**

- A larger footprint and slower angular motion lengthen an overhead pass. These examples give 4.5 and 9.3 min above 25°.

► **A pass need not go overhead.**

- An off-center pass usually has a lower peak and shorter contact. Some passes never reach the required elevation.

Two ideal overhead passes, one mask



At 25°: **4.5 min** at 550 km, **9.3 min** at 1200 km
Ideal duration $\Delta t = 2\psi/\omega$, where $\omega = 2\pi/T$

Circular overhead passes with Earth rotation omitted. ψ is the footprint center angle. Actual contacts depend on relative motion.

Coverage Over Time Is the Union of Contacts

A constellation offers access whenever at least one satellite meets the terminal's access conditions.

► **Combine the individual windows.**

- Let $N(t)$ be the number of usable satellites. A terminal has geometric access when $N(t) \geq 1$. A gap occurs when $N(t) = 0$.

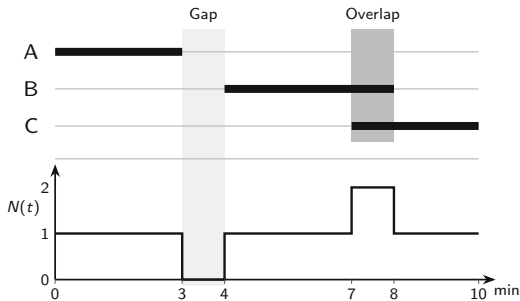
► **Overlap adds possible alternatives.**

- Here B and C overlap for one minute. Two candidates offer a choice, but do not automatically give twice the capacity.

► **An average can hide an interruption.**

- The mean count is one, yet access stops for a minute. Report coverage and gaps alongside the mean.

A mean count of 1 does not mean no gaps



Covered: $9/10 = 90\%$ Mean count: $10/10 = 1$

Constructed 10-minute example: A [0,3), B [4,8), C [7,10]. Geometric access is not proof of an end-to-end Internet route.

More Satellites Cannot Fix the Wrong Latitude Reach

Check coverage at each location that needs service. A large constellation can still miss a region.

► Combine orbital tilt and footprint.

- A 53° orbit reaches 53° N below the satellite. At 550 km and a 25° mask, its footprint extends at most another 8.46° .

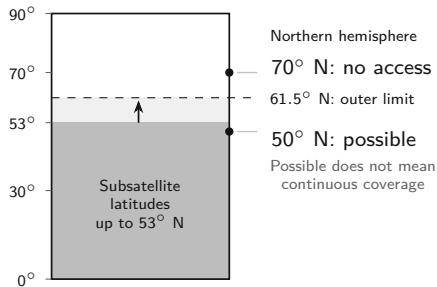
► Some locations are out of reach.

- The northern limit is about 61.5° N. Adding more satellites to this same shell cannot provide access at 70° N.

► Being inside the limit is not enough.

- A terminal at 50° N is geographically eligible. Its actual contact times still depend on longitude, spacing, and phase.

A necessary condition for northern access



$$\text{Northern limit} \leq \min(90^\circ, i + \psi)$$

Ideal 53° circular shell, 550 km altitude, 25° elevation mask. A geometric limit, not a guarantee of service inside it.

Test the Prediction by Changing Only Inclination

Use SatNet Edu to compare the same constellation at two inclinations and two ground latitudes.

► Predict before running.

- The 53° shell cannot reach the 70° N terminal with these settings. An 85° shell makes access possible, not guaranteed.

► Keep the other variables fixed.

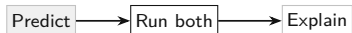
- Use the same altitude, satellite count, longitude, epoch, mask, and sampling interval. Change only inclination.

► Inspect access at both locations.

- Count usable ground links at each sample. Compare covered-sample fractions and gaps. Explain any tradeoff at 50° N.

A controlled two-scenario comparison

Setting	Scenario A	Scenario B
Inclination	53°	85°
Altitude	550 km	550 km
Planes $\times$ satellites	6×12	6×12
Minimum elevation	25°	25°
Ground terminals	50° N and 70° N, both 0° E	
Duration / step	6 hours / 30 seconds	



$$\text{Covered-sample fraction} = \frac{\text{samples with at least one usable ground link}}{\text{all recorded samples}}$$

Use a common epoch and no failures. Sampled access can miss short gaps and does not establish an end-to-end route.